INCH-POUND

MIL-M-28787/251
AMENDMENT 3
31 December 1992
SUPERSEDING
AMENDMENT 2
25 October 1985

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

LATCH, DIGITAL, KEY CODE AMG

This amendment forms a part of Military Specification MIL-M-28787/251(EC), dated 24 June 1982, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGES 4, 5, 6, 7, 8, 9, AND 10

*TABLE I: Delete and substitute new table I as printed on pages 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, and 14 of this amendment.

PAGE 11

*4.4.2: Delete and substitute the following:

"4.4.2 Order of testing. Test conditions A through FF shall be performed in any order with the following exceptions. Test conditions A, B, and C are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, and C where applicable, followed by the remaining tests in the sequence."

*4.5.1, lines 5 and 6: Delete "B through K and N through DD" and substitute "D through DD".

4.5.1, eighth line: Delete "100°C" and substitute "85°C".

*4.5.3, line 3: Delete "B through DD" and substitute "D through DD".

*4.5.3, line 6: Delete "B through FF" and substitute "D through FF".

6.4, second line: Delete "100°C" and substitute "85°C".

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PAGE 12

*6.4, Input current: Delete "18 V" (two places) and substitute "15 V" (two places).

6.4, Output drive current: Delete "100°C" (two places) and substitute "85°C" (two places).

PAGE 14

*FIGURE 2, ITEM P07: Delete "18 V" and substitute "15 V".

PAGE 15

*FIGURE 3: Delete and substitute new figure 3 as printed on page 15 of this amendment.

PAGE 16

*FIGURE 4, ITEM P07: Delete "18 V" and substitute "15 V".

PAGE 33

FIGURE 9, NOTE 2: Delete "100°C" and substitute "85°C".

PAGE 35

FIGURE 10, NOTE 2: Delete "100°C" and substitute "85°C".

PAGE 41

*FIGURE 12, ITEM P07: Delete "18 V" and substitute "15 V".

PAGE 44

*FIGURE 13, ITEM P07: Delete "18 V" and substitute "15 V".

PAGES 82, 83, 84 AND 85

TABLE XXV, Test conditions Y41-Y80, Connect A to pin column: Delete "3, 11, 36" (12 places).

TABLE XXV, Test conditions Y41-Y80, Connect B to pin column: Add "3, 11, 36" (40 places).

TABLE XXV, Test conditions Y53-Y60, Connect A to pin column: Add an arrow between Y53 and Y60.

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PAGE 87

TABLE XXV, Test conditions Y91-Y100, Parameter column: Delete "low" and substitute "high".

PAGE 89

TABLE XXV, Test conditions Y111-Y120, Parameter column: Delete "low" and substitute "high".

NOTE: The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - NW

(Project 5963-0203)

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TABLE I. Module electrical requirements.

Test cond	Parameter	Ref test table	Test type 1/-3/	25°C ± 5°C 4 /		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	Q	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B18	Input clamp diode voltage, negative	IV	S	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	V
C1-C18	Input clamp diode voltage, positive	V	S	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	V
D1-D64	Functional	VI	X	See table VI										
E1-E20	Output voltage, high level (IOH = 0)	VII	X	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	V
F1-F20	Output voltage, low level (IOL = 0)	VIII	X	0	50	0	50	0	50	0	50	0	50	mV
G1-G20	Output voltage, high level (VDD = 5 V)	IX	X	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	V
H1-H20	Output voltage, low level (VDD = 5 V)	X	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
J1-J20	Output voltage, high level (VDD = 15 V)	XI	X	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	V

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type 1/-3/	25°C ± 5°C 4 /		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
K1-K20	Output voltage, low level (VDD = 15 V)	XII	X	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	V
L1-L18	Input current, high level	XIII	X	0	200	---	---	0	200	---	---	0	200	nA
M1-M18	Input current, low level	XIV	X	0	-200	---	---	0	-200	---	---	0	-200	nA
N1	Power supply current, low level (Q outputs)	XV	X	0	60	0	60	0	600	0	60	0	975	µA
N2	Power supply current, high level (Q outputs)	XV	X	0	60	0	60	0	600	0	60	0	975	µA
P1-P20	Transition time (tTLH), CL = 50 pF, VDD = 5 V	XVI	S	36	360	36	360	40	400	27	275	43	425	ns
P21-P40	Transition time (tTLH), CL = 150 pF, VDD = 5 V	XVI	Q	84	840	84	840	93	930	64	640	99	993	ns
Q1-Q20	Transition time (tTHL), CL = 50 pF, VDD = 5 V	XVII	S	20	200	20	200	22	220	15	150	23	236	ns
Q21-Q40	Transition time (tTHL), CL = 150 pF, VDD = 5 V	XVII	Q	68	680	68	680	75	750	51	515	80	803	ns
R1-R20	Transition time (tTLH), CL = 50 pF, VDD = 10 V	XVIII	Q	18	180	18	180	20	200	13	135	21	213	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type 1/-3/	25°C ± 5°C 4 /		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
S1-S20	Transition time (tTHL), CL = 50 pF, VDD = 10 V	XIX	Q	10	100	10	100	11	110	7	75	12	119	ns
T1-T20	Transition time (tTLH), CL = 50 pF, VDD = 15 V	XX	Q	13	130	13	130	14	145	10	100	15	154	ns
U1-U20	Transition time (tTHL), CL = 50 pF, VDD = 15 V	XXI	Q	8	80	8	80	9	90	6	60	10	96	ns
V1-V20	Propagation delay time (tPLH), CL = 50 pF, VDD = 5 V	XXII	S	44	440	44	440	48	485	33	335	52	519	ns
V21-V40	Propagation delay time (tPHL), CL = 50 pF, VDD = 5 V	XXII	S	44	440	44	440	48	485	33	335	52	519	ns
V41-V60	Propagation delay time (tPLH), CL = 150 pF, VDD = 5 V	XXII	Q	64	640	64	640	70	705	48	485	75	755	ns
V61-V80	Propagation delay time (tPHL), CL = 150 pF, VDD = 5 V	XXII	Q	64	640	64	640	70	705	48	485	75	755	ns
W1-W20	Propagation delay time (tPLH), CL = 50 pF, VDD = 10 V	XXII	Q	18	180	18	180	20	200	13	135	21	213	ns
W21-W40	Propagation delay time (tPHL), CL = 50 pF, VDD = 10 V	XXIII	Q	18	180	18	180	20	200	13	135	21	213	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type 1/-3/	25°C ± 5°C 4 /		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
X1-X20	Propagation delay time (tPLH), CL = 50 pF, VDD = 15 V	XXIV	Q	12	120	12	120	13	135	9	90	14	141	ns
X21-X40	Propagation delay time (tPHL), CL = 50 pF, VDD = 15 V	XXIV	Q	12	120	12	120	13	135	9	90	14	141	ns
Y1,Y3,Y5, Y7,Y9-Y13, Y15,Y17, Y19	Propagation delay time (tPLH), CL = 50 pF, VDD = 5 V, polarity high	XXV	S	45	450	45	450	50	500	34	340	53	531	ns
Y2,Y4,Y6, Y8,Y14, Y16,Y18, Y20	Propagation delay time (tPLH), CL = 50 pF, VDD = 5 V, polarity high	XXV	S	50	500	50	500	55	555	38	380	58	591	ns
Y21,Y23, Y25,Y27, Y29-Y33, Y35,Y37, Y39	Propagation delay time (tPHL), CL = 50 pF, VDD = 5 V, polarity high	XXV	S	45	450	45	450	50	500	34	340	53	531	ns
Y22,Y24, Y26,Y28, Y34,Y36, Y38,Y40	Propagation delay time (tPHL), CL = 50 pF, VDD = 5 V, polarity high	XXV	S	50	500	50	500	55	550	38	380	58	591	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type 1/-3/	25°C ± 5°C 4 /		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Y41,Y43, Y45,Y47, Y49-Y53, Y55,Y57, Y59	Propagation delay time (tPLH), CL = 50 pF, VDD = 5 V, polarity low	XXV	S	45	450	45	450	50	500	34	340	53	531	ns
Y42,Y44, Y46,Y48, Y54,Y56, Y58,Y60	Propagation delay time (tPLH), CL = 50 pF, VDD = 5 V, polarity low	XXV	S	50	500	50	500	55	550	38	380	58	591	ns
Y61,Y63, Y65,Y67, Y69-Y73, Y75,Y77, Y79	Propagation delay time (tPHL), CL = 50 pF, VDD = 5 V, polarity low	XXV	S	45	450	45	450	50	500	34	340	53	531	ns
Y62,Y64, Y66,Y68, Y74,Y76, Y78,Y80	Propagation delay time (tPHL), CL = 50 pF, VDD = 5 V, polarity low	XXV	S	50	500	50	500	55	550	38	380	58	591	ns
Y81,Y83, Y85,Y87, Y89-Y93, Y95,Y97, Y99	Propagation delay time (tPLH), CL = 150 pF, VDD = 5 V, polarity high	XXV	Q	65	650	65	650	72	720	49	495	76	767	ns

See footnotes at end of table.

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Test cond	Parameter	Ref test table	Test type 1/-3/	25°C ± 5°C 4 /		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Y82,Y84, Y86,Y88, Y94,Y96, Y98,Y100	Propagation delay time (tPLH), CL = 150 pF, VDD = 5 V, polarity high	XXV	Q	70	700	70	700	77	775	53	530	83	828	ns
Y101,Y103, Y105,Y107, Y109-Y113, Y115,Y117, Y119	Propagation delay time (tPHL), CL = 150 pF, VDD = 5 V, polarity high	XXV	Q	65	650	65	650	72	720	49	495	76	767	ns
Y102,Y104, Y106,Y108, Y114,Y116, Y118,Y120	Propagation delay time (tPHL), CL = 150 pF, VDD = 5 V, polarity high	XXV	Q	70	700	70	700	77	775	53	530	83	828	ns
Y121,Y123, Y125,Y127, Y129-Y133, Y135,Y137, Y139	Propagation delay time (tPLH), CL = 150 pF, VDD = 5 V, polarity low	XXV	Q	65	650	65	650	72	720	49	495	76	767	ns
Y122,Y124, Y126,Y128, Y134,Y136, Y138,Y140	Propagation delay time (tPLH), CL = 150 pF, VDD = 5 V, polarity low	XXV	Q	70	700	70	700	77	775	53	530	83	828	ns

See footnotes at end of table.

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						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Y141,Y143, Y145,Y147, Y149-Y153, Y155,Y157, Y159	Propagation delay time (tPHL), CL = 150 pF, VDD = 5 V, polarity low	XXV	Q	65	650	65	650	72	720	49	495	76	767	ns
Y142,Y144, Y146,Y148, Y154,Y156, Y158,Y160	Propagation delay time (tPHL), CL = 150 pF, VDD = 5 V, polarity low	XXV	Q	70	700	70	700	77	775	53	530	83	828	ns
Z1,Z3,Z5, Z7,Z9-Z13, Z15,Z17, Z19	Propagation delay time (tPLH), CL = 50 pF, VDD = 10 V, polarity high	XXVI	Q	20	200	20	200	21	215	16	160	24	237	ns
Z2,Z4,Z6, Z8,Z14, Z16,Z18, Z20	Propagation delay time (tPLH), CL = 50 pF, VDD = 10 V, polarity high	XXVI	Q	23	230	23	230	25	250	18	185	26	263	ns
Z21,Z23, Z25,Z27, Z29-Z33, Z35,Z37, Z39	Propagation delay time (tPHL), CL = 50 pF, VDD = 10 V, polarity high	XXVI	Q	20	200	20	200	21	215	16	160	24	237	ns

See footnotes at end of table.

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						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Z22,Z24, Z26,Z28, Z34,Z36, Z38,Z40	Propagation delay time (tPHL), CL = 50 pF, VDD = 10 V, polarity high	XXVI	Q	23	230	23	230	25	250	18	185	26	263	ns
Z41,Z43, Z45,Z47, Z49-Z53, Z55,Z57, Z59	Propagation delay time (tPLH), CL = 50 pF, VDD = 10 V, polarity low	XXVI	Q	20	200	20	200	21	215	16	160	24	237	ns
Z42,Z44, Z46,Z48, Z54,Z56, Z58,Z60	Propagation delay time (tPLH), CL = 50 pF, VDD = 10 V, polarity low	XXVI	Q	23	230	23	230	25	250	18	185	26	263	ns
Z61,Z63, Z65,Z67, Z69-Z73, Z75,Z77, Z79	Propagation delay time (tPHL), CL = 50 pF, VDD = 10 V, polarity low	XXVI	Q	20	200	20	200	21	215	16	160	24	237	ns
Z62,Z64, Z66,Z68, Z74,Z76, Z78,Z80	Propagation delay time (tPHL), CL = 50 pF, VDD = 10 V, polarity low	XXVI	Q	23	230	23	230	25	250	18	185	26	263	ns

See footnotes at end of table.

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						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
AA1,AA3, AA5,AA7, AA9-AA13, AA15,AA17, AA19	Propagation delay time (tPLH), CL = 50 pF, VDD = 15 V, polarity high	XXVII	Q	16	160	16	160	17	175	12	120	20	197	ns
AA2,AA4, AA6,AA8, AA14,AA16, AA18,AA20	Propagation delay time (tPLH), CL = 50 pF, VDD = 15 V, polarity high	XXVII	Q	18	180	18	180	20	200	13	135	21	213	ns
AA21,AA23, AA25,AA27, AA29-AA33, AA35,AA37, AA39	Propagation delay time (tPHL), CL = 50 pF, VDD = 15 V, polarity high	XXVII	Q	16	160	16	160	17	175	12	120	20	197	ns
AA22,AA24, AA26,AA28, AA34,AA36, AA38,AA40	Propagation delay time (tPHL), CL = 50 pF, VDD = 15 V, polarity high	XXVII	Q	18	180	18	180	20	200	13	135	21	213	ns
AA41,AA43, AA45,AA47, AA49-AA53, AA55,AA57, AA59	Propagation delay time (tPLH), CL = 50 pF, VDD = 15 V, polarity low	XXVII	Q	16	160	16	160	17	175	12	120	20	197	ns

See footnotes at end of table.

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Test cond	Parameter	Ref test table	Test type 1/-3/	25°C ± 5°C 4 /		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
AA42,AA44, AA46,AA48, AA54,AA56, AA58,AA60	Propagation delay time (tPLH), CL = 50 pF, VDD = 15 V, polarity low	XXVII	Q	18	180	18	180	20	200	13	135	21	213	ns
AA61,AA63, AA65,AA67, AA69-AA73, AA75,AA77, AA79	Propagation delay time (tPHL), CL = 50 pF, VDD = 15 V, polarity low	XXVII	Q	16	160	16	160	17	175	12	120	20	197	ns
AA62,AA64, AA66,AA68, AA74,AA76, AA78,AA80	Propagation delay time (tPHL), CL = 50 pF, VDD = 15 V, polarity low	XXVII	Q	18	180	18	180	20	200	13	135	21	213	ns
BB1-BB20	Data hold time check, VDD = 5 V	XXVIII	S	1560	164	1560	1640	1560	1640	1560	1640	1560	1640	ns
CC1-CC20	Data hold time check, VDD = 10 V	XXIX	Q	1560	164	1560	1640	1560	1640	1560	1640	1560	1640	ns
DD1-DD20	Data hold time check, VDD = 15 V	XXX	Q	1560	164	1560	1640	1560	1640	1560	1640	1560	1640	ns

See footnotes at end of table.

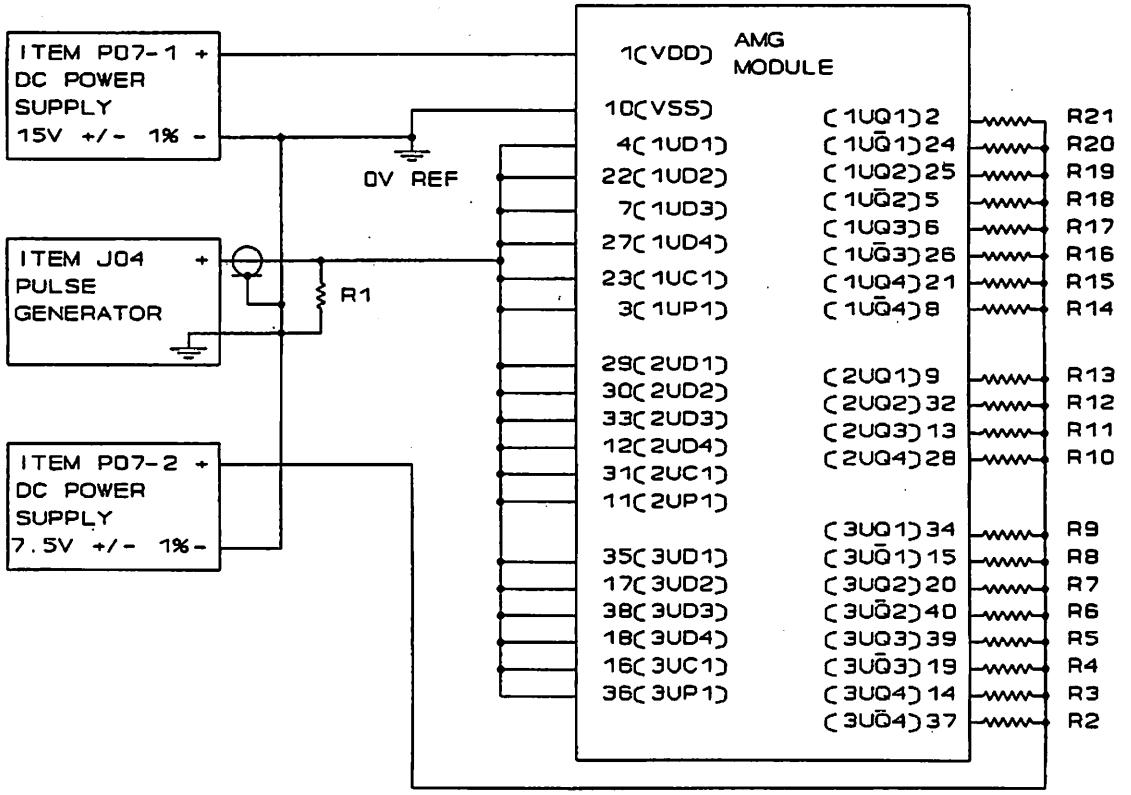
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						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
EE	Power supply filter capacitance	XXXI	S	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	μF
FF1-FF18	Input capacitance	XXXII	Q	2	15	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

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NOTES:

1. R1 = 51.1Ω ± 5%, 5 W.
2. Item J04 setup:
   PA = 15 V ± 1%.
   PW = 5 ± 0.5 uS.
   tTLH = tTHL = 50 ± 10 ns.
   Frequency = 100 KHz ± 20%.
3. R2-R20 = 2.49 kΩ ± 1%, 1/4 W.

FIGURE 3. Component temperature test circuit.

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